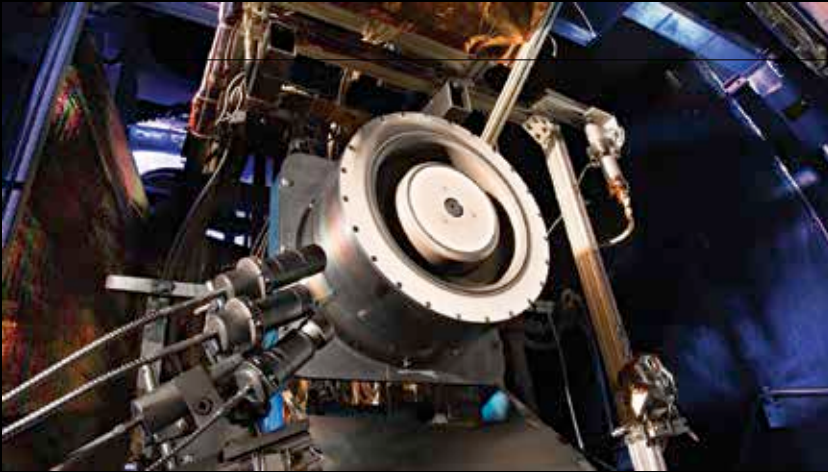




What Powers A Spacecraft?

Solar Energy, Batteries, Atoms... But how does it all work?

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The ability to change one's state or position without energy is an impossible task, and just like any other machine, spacecraft also require power to function. The problem, however, lies in where the spacecraft gets its power. There are several ways to generate power when you are on Earth, like combustion, harvesting the natural forces like wind, the sea, flowing water like in the rivers or even capture some solar energy directly thanks to solar panels.

A spacecraft though, usually gets its power from one of three sources; the Sun, batteries, or a radioisotope. The power system for a spacecraft mostly depends on the kind of mission and what the engineers expect. These spacecraft already have the instruments to take

pictures and collect information; all they need now is a power source to get them to the right location and send this information back to Earth.

POWERING A SPACECRAFT USING THE SUN

The Sun is one of the most convenient, clean and powerful sources of energy to exist, especially for a spacecraft. The only thing left is to convert this solar energy into usable electrical power. The spacecraft that orbit the Earth are high enough that they are out of the reach of Earth's atmosphere and can obtain enough energy from the Sun to power themselves. The solar

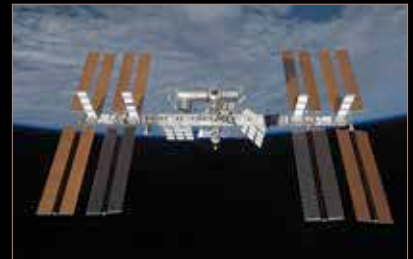


panels present in these spacecraft convert the Sun's energy into electrical power, which helps the spacecraft function.

HOW DO SOLAR CELLS WORK?

The photovoltaic cells that are found in solar panels are made of semiconductive material, primarily silicon, which is capable of absorbing light from the Sun and converting it into electricity to power the spacecraft. The electricity from the solar panels is used to charge the batteries in a spacecraft. These batteries help power the spacecraft when it doesn't get access to direct sunlight.

Real photovoltaic cells are designed with the basic materials, doping, and junction shapes in mind to maximize their ability to convert light energy into electrical energy. Each cell is capable of producing a small amount of current at a low voltage, similar to a standard pen-light battery. Many of them must be connected in series to generate the amount of electric power required for a satellite to operate and meet the power demands of its on-board instruments.



WHY ARE SOLAR ARRAYS SO LARGE?

The strength of the Sun doesn't matter when you are powering even an average-sized spacecraft, as the solar arrays required are huge. This mostly has to do with the fact that individual solar cells have relatively low efficiency.

This is why most images of traditional satellites show a pair of long wings extending from their sides, which serve as 'solar panels.' Solar cells based on semiconductor materials such as Gallium-arsenide/Arsenicum are now available, with efficiency figures nearly double those of silicon cells. These new types of cells will enable future space missions to use smaller solar